lum lookup tables

Three lookup files supply numeric parameters that `landuse.lum` references by name: `cntable.lum` (SCS curve numbers), `cons_practice.lum` (USLE P factor and maximum slope length), and `ovn_table.lum` (overland-flow Manning's n). Each file has the same shape: a title line, a header line, then one record per row. Rows are read with list-directed I/O, so any column after the last typed field is ignored by the solver.

cntable.lum

Curve number table. Each entry holds four SCS-CN values for hydrologic soil groups A, B, C, and D.

Source

- Reader: `src/cntbl_read.f90`
- Type: `src/landuse_data_module.f90`, `type curvenumber_table` (instances stored in `cn`)
- Sized into `db_mx%cn_lu` after the file is read.

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- One record per curve number entry. Fields are list-directed.

#	Field	Type	Default	Description
1	name	char(40)	blank	curve number entry name. Referenced from `landuse.lum` field `cn_lu`
2	cn(1)	real	30.0	curve number for hydrologic soil group A
3	cn(2)	real	55.0	curve number for hydrologic soil group B
4	cn(3)	real	70.0	curve number for hydrologic soil group C
5	cn(4)	real	77.0	curve number for hydrologic soil group D

Curve numbers are dimensionless and bounded by 0 and 100. SWAT+ Editor writes additional `description`, `treat`, and `cond_cov` columns after `cn(4)`. The reader does not consume them.

Example

Excerpt from `refdata/Ames_sub1/cntable.lum`:

```
cntable.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                      cn_a          cn_b          cn_c          cn_d  description
fal_bare               77.00000      86.00000      91.00000      94.00000  Fallow
rc_strow_g             67.00000      78.00000      85.00000      89.00000  Row_crops
wood_f                 36.00000      60.00000      73.00000      79.00000  Woods
urban                  98.00000      98.00000      98.00000      98.00000  Paved_parking_lots_roofs_drivewa
```

cons_practice.lum

Conservation practice table. Supplies the USLE support practice factor (P) and the maximum slope length used for length-slope factor calculations.

Source

- Reader: src/cons_prac_read.f90
- Type: src/landuse_data_module.f90, `type conservation_practice_table` (instances stored in `cons_prac`)
- Sized into `db_mx%cons_prac` after the file is read.

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- One record per practice. Fields are list-directed.

#	Field	Type	Default	Units	Description
1	name	char(40)	blank	-	practice name. Referenced from `landuse.lum` field `cons_prac`
2	pfac	real	1.0	-	USLE support practice factor (P). Dimensionless, 0 to 1
3	sl_len_mx	real	1.0	m	maximum slope length for the practice

SWAT+ Editor writes a `description` column after `sl_len_mx`. The reader does not consume it.

Example

Excerpt from `refdata/Ames_sub1/cons_practice.lum`:

```
cons_practice.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                    usle_p   slp_len_max  description
up_down_slope          1.00000     121.00000  Up_and_down_slope
cross_slope            0.75000     121.00000  Cross_slope_tillage
contour_farming        0.50000     121.00000  Contour_tillage
strip_contour          0.25000     121.00000  Strip_cropping_contour
ter_3-8_sodout         0.50000      76.00000  terraces_3-8%_slopes_sod-outlet
```

ovn_table.lum

Overland-flow Manning's n table. Supplies a mean, minimum, and maximum n per land use class for sheet-flow routing across the HRU.

Source

- Reader: src/overland_n_read.f90
- Type: src/landuse_data_module.f90, `type overlandflow_n_table` (instances stored in `overland_n`)
- Sized into `db_mx%ovn` after the file is read.

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- One record per class. Fields are list-directed.

#	Field	Type	Default	Units	Description
1	name	char(40)	blank	-	class name. Referenced from `landuse.lum` field `ovn`
2	ovn	real	0.5	s m^(-1/3)	mean Manning's n for overland flow
3	ovn_min	real	0.5	s m^(-1/3)	minimum Manning's n for overland flow
4	ovn_max	real	0.5	s m^(-1/3)	maximum Manning's n for overland flow

SWAT+ Editor writes a `description` column after `ovn_max`. The reader does not consume it.

Example

Excerpt from `refdata/Ames_sub1/ovn_table.lum`:

```
ovn_table.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                  ovn_mean       ovn_min       ovn_max  description
convtill_nores         0.09000       0.06000       0.12000  Conventional_tillage_no_residue
convtill_res           0.19000       0.16000       0.22000  Conventional_tillage_residue
notill_2-9res          0.30000       0.17000       0.47000  No_till_2-9_t/ha_residue
forest_med             0.60000       0.50000       0.70000  Forest_medimum_good
urban_asphalt          0.01100       0.01100       0.01100  Urban_asphalt
```

Related files

- `landuse.lum`. Master land-use management table. Its `cn_lu`, `cons_prac`, and `ovn` columns name rows in these three files.
- `hru-data.hru`. Selects a `landuse.lum` row per HRU, which in turn selects rows in these three tables.